



Fatigue Design of Mechanical Joints and Aerospatial Structures (PFLMEA)

Collection of Exercises

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⚡ Interactive Computing using Jupyter Notebook

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📋 **AI-use disclosure.** Parts of this booklet (LaTeX scaffolding, code drafts, and formatting) were prepared with the assistance of *Claude Code* (Anthropic). All content was reviewed, verified, and edited by the author, who takes full responsibility for it.

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1 Problems 1

Formulary

The following definitions and relations are used throughout this exercise sheet. Consider a constant-amplitude cyclic stress history characterised by a sinusoidal waveform.

Maximum stress:	$\sigma_{\max}$	Minimum stress:	$\sigma_{\min}$
Mean stress:	$\sigma_m = \frac{\sigma_{\max} + \sigma_{\min}}{2}$	Stress amplitude:	$\sigma_a = \frac{\sigma_{\max} - \sigma_{\min}}{2}$
Stress range:	$\Delta\sigma = \sigma_{\max} - \sigma_{\min} = 2\sigma_a$	Stress ratio:	$R = \frac{\sigma_{\min}}{\sigma_{\max}}$
Amplitude ratio:	$A = \frac{\sigma_a}{\sigma_m} = \frac{1 - R}{1 + R}$		

S–N curve (power-law form):

$$\Delta\sigma^m N_f = C \quad (\text{constant})$$

Basquin relation:

$$\frac{\Delta\sigma}{2} = \sigma_a = \sigma'_f (2N_f)^b$$

where σ'_f is the fatigue strength coefficient and b is the fatigue strength exponent.

Problem 1.

Characterisation of a General Constant-Amplitude Cycle

Consider the constant-amplitude cyclic loading depicted in Figure 1.1, characterised by a sinusoidal stress waveform. The loading has the potential to induce fatigue damage in a given material if applied for a sufficient number of cycles, even though the maximum stress remains below the static strength of the material (*e.g.* below the yield stress). Only one cycle ($N = 1$) is shown in the figure.

From the graph, the following values can be read:

$$\sigma_{\max} = 150 \text{ MPa}, \quad \sigma_{\min} = -50 \text{ MPa}.$$

- Compute the mean stress σ_m .
- Compute the stress amplitude σ_a .
- Compute the stress range $\Delta\sigma$.
- Compute the stress ratio R .
- Compute the amplitude ratio A and verify the relation $A = (1 - R)/(1 + R)$.

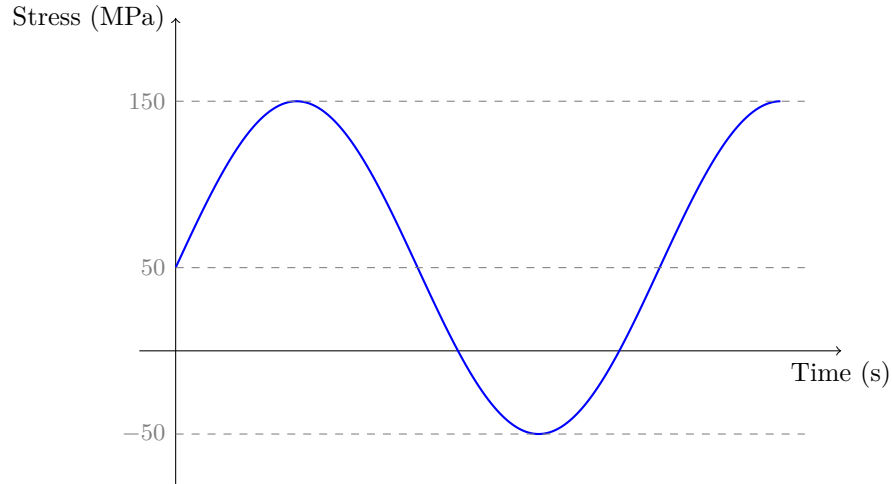


Figure 1.1: Constant-amplitude cyclic loading with $\sigma_{\max} = 150$ MPa and $\sigma_{\min} = -50$ MPa.

- (f) Classify this loading: is it pulsating ($R = 0$), fully reversed ($R = -1$), or general ($R \neq 0, -1$)?

Remark

Only two independent parameters from the set $\{\sigma_{\max}, \sigma_{\min}, \sigma_m, \sigma_a, \Delta\sigma, R\}$ are needed to fully define the constant-amplitude cycle. The remaining parameters are derived from that pair. The pair (R, A) cannot be used on its own, because A is a function of R alone and carries no additional information about the stress magnitude.

Problem 2.

Pulsating constant-amplitude loading

Consider the constant-amplitude pulsating loading shown in Figure 1.2. From the graph, the following peak values can be read:

$$\sigma_{\max} = 200 \text{ MPa}, \quad \sigma_{\min} = 0 \text{ MPa}.$$

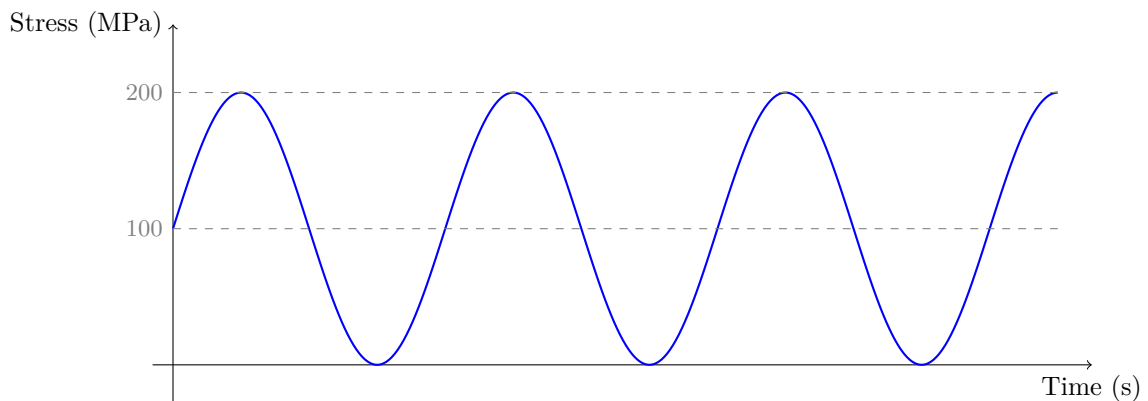


Figure 1.2: Pulsating constant-amplitude loading ($\sigma_{\min} = 0$).

- (a) Compute all cyclic stress parameters: σ_m , σ_a , $\Delta\sigma$, R , and A .
 (b) Verify that for pulsating loading: $R = 0$, $A = 1$, $\sigma_a = \sigma_m$, and $\Delta\sigma = \sigma_{\max}$.

Problem 3.**Fully reversed constant-amplitude loading**

Consider the constant-amplitude fully reversed (alternating) loading shown in Figure 1.3. From the graph, the following peak values can be read:

$$\sigma_{\max} = 200 \text{ MPa}, \quad \sigma_{\min} = -200 \text{ MPa}.$$

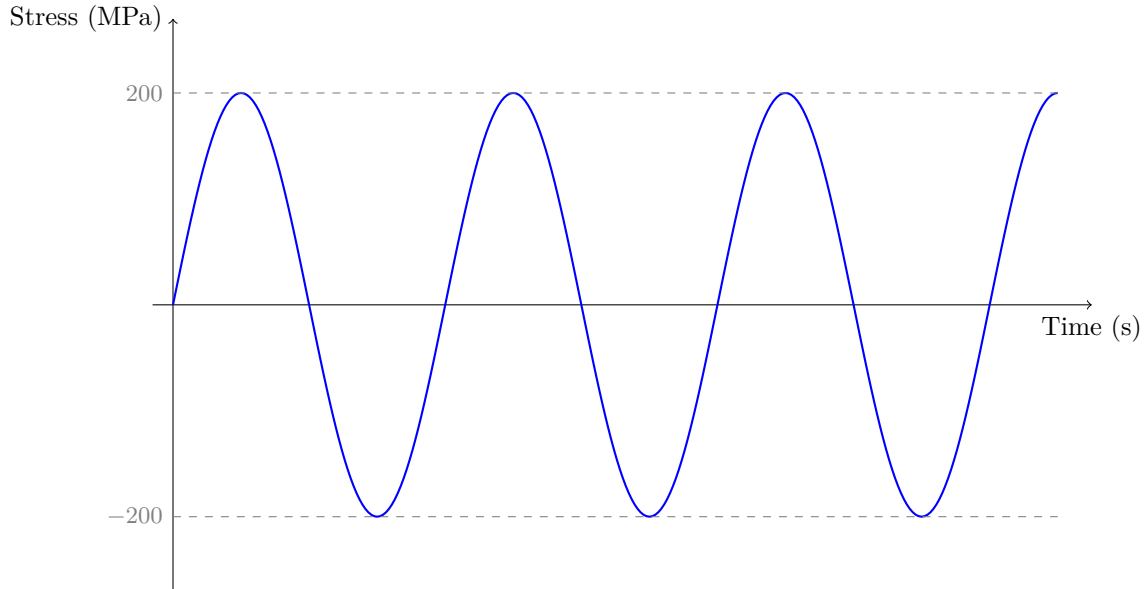


Figure 1.3: Fully reversed constant-amplitude loading ($\sigma_{\min} = -\sigma_{\max}$).

- (a) Compute all cyclic stress parameters: σ_m , σ_a , $\Delta\sigma$, R , and A .
- (b) Verify that for fully reversed loading: $R = -1$, $\sigma_m = 0$, and $\sigma_a = \sigma_{\max}$.
- (c) Show that $A \rightarrow \infty$ when $R \rightarrow -1$. Explain why the pair (σ_m, R) with $R = -1$ is insufficient to define the loading uniquely.

Problem 4.**General relations between cyclic stress parameters**

Consider a constant-amplitude tensile stress cycle for which the stress ratio R and the mean stress σ_m are known.

- (a) Starting from the definitions of σ_m and R , set up a system of two equations in the two unknowns $\sigma_{\max}$ and $\sigma_{\min}$. Solve the system to obtain:

$$\sigma_{\max} = \frac{2\sigma_m}{1+R}, \quad \sigma_{\min} = \frac{2R\sigma_m}{1+R}.$$

- (b) Derive the stress amplitude and stress range in terms of σ_m and R :

$$\sigma_a = \frac{1-R}{1+R}\sigma_m, \quad \Delta\sigma = \frac{2(1-R)}{1+R}\sigma_m.$$

- (c) Confirm that $A = \sigma_a/\sigma_m = (1-R)/(1+R)$, which is consistent with the earlier definition.
- (d) Explain why, for $R = -1$, the parameters σ_m and R are not sufficient to define the loading cycle. How many distinct fully reversed cycles share the same $\sigma_m = 0$ and $R = -1$?

Problem 5.

S–N curve of a structural steel (Basquin law)

Consider an S–N curve representative of the fatigue behaviour of a structural steel detail, extracted from Eurocode 3. In the finite-life region, the curve is described by the following power-law relation:

$$\Delta\sigma^m N_f = C,$$

where C is a constant. Assume $m = 3$ and that $N_f = 2 \times 10^6$ cycles when $\Delta\sigma = 160$ MPa (this defines the detail category FAT 160).

- (a) Compute the constant C .
- (b) Rewrite the S–N relation in the Basquin form:

$$\frac{\Delta\sigma}{2} = \sigma'_f (2N_f)^b.$$

Determine the fatigue strength coefficient σ'_f and the fatigue strength exponent b .

Hint: Starting from $\Delta\sigma^m N_f = C$, solve for $\Delta\sigma$ as a function of N_f and rearrange to match the Basquin form.

- (c) Compute the fatigue life N_f for a stress range $\Delta\sigma = 160$ MPa.
- (d) Compute the fatigue strength (*i.e.* the stress range $\Delta\sigma$) for a life of $N_f = 1 \times 10^6$ cycles.
- (e) The fatigue limit is defined as the stress range corresponding to a life of 5×10^6 cycles (Eurocode 3 criterion). Compute the fatigue limit $\Delta\sigma_D$.
- (f) Consider an applied stress range of $\Delta\sigma = 300$ MPa.
 - a) Compute the fatigue life N_f .
 - b) Now reduce the stress range by 10 % to $\Delta\sigma = 270$ MPa and recompute the fatigue life.
 - c) Express the life increase as a percentage.
 - d) Comment on the non-linearity of the S–N relation: why does a modest reduction in stress range lead to a disproportionate increase in fatigue life?

Problem 6.

Conceptual questions on fatigue fundamentals

The following conceptual questions are drawn from Schijve (2009), Chapter 2. They should be discussed in class and do not require numerical calculations.

- (a) The fatigue life of a component is conventionally divided into three phases: crack initiation, crack growth, and final failure.
 - i) Which factors are significant for each of these three phases?
 - ii) What is the most essential feature of the crack initiation period?
 - iii) What is an important difference between the crack initiation and crack growth periods?

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- iv) What is an important material characteristic governing the crack growth period?
- (b) Give two different definitions of the fatigue limit S_f . Under what circumstances may the fatigue limit not exist as a well-defined horizontal asymptote of the S–N curve?
- (c) What is the difference between high-cycle fatigue (HCF) and low-cycle fatigue (LCF)?
- (d) An S–N curve has an upper and a lower horizontal asymptote. Explain the physical meaning of each.
- (e) On a double-logarithmic scale, an S–N curve often exhibits a large linear part. Which equation describes this part, and what are the roles of its constants?
- (f) The fatigue limit for $S_m = 0$ within a particular alloy family (*e.g.* steels, aluminium alloys) is often found to be linearly related to the ultimate tensile strength of the material. This implies that a stronger material should have a higher fatigue limit.
- i) Is this a useful trend for material selection? Discuss limitations.
- ii) The linear relation between S_f and S_u for steels is sometimes not maintained for very high-strength steels. Explain why.

Answer:  [Open in Colab](#)

2 Problems 2

Formulary

Stress concentration factor (based on net-section nominal stress):

$$K_t = \frac{\sigma_{\text{peak}}}{\sigma_{\text{nom}}} \quad (2.1)$$

where σ_{peak} is the maximum (elastic) stress at the notch root and σ_{nom} is the nominal stress on the net section.

Creager–Paris solution — maximum elastic stress at the tip of a crack of half-length a with a finite root radius ρ in an infinite plate under remote uniform stress σ_{∞} :

$$\sigma_m = \sigma_{\infty} \left(1 + 2\sqrt{\frac{a}{\rho}} \right) \quad (2.2)$$

Special cases: (i) $a = 0$ (no defect) $\Rightarrow \sigma_m = \sigma_{\infty}$, hence $K_t = 1$; (ii) $a = \rho$ (circular hole) $\Rightarrow K_t = 3$; (iii) $\rho \rightarrow 0$ (sharp crack) $\Rightarrow \sigma_m \rightarrow \infty$, motivating the use of fracture mechanics.

Approximate form for a sharp defect ($a \gg \rho$):

$$\sigma_m \approx 2\sigma_{\infty}\sqrt{\frac{a}{\rho}} \quad (2.3)$$

Elliptical hole in an infinite plate under uniaxial tension (Inglis, 1913):

$$K_t = 1 + 2\frac{a}{b} \quad (2.4)$$

where a is the semi-axis perpendicular to the loading direction and b is the semi-axis parallel to it. Using $\rho = b^2/a$ (tip radius of the ellipse), this recovers Eq. (2.2).

Circular hole in a finite-width strip (width W , hole diameter D):

$$K_{tg} \approx 3 - 3.14\left(\frac{D}{W}\right) + 3.667\left(\frac{D}{W}\right)^2 - 1.527\left(\frac{D}{W}\right)^3 \quad (2.5)$$

where $K_{tg} = \sigma_{\text{peak}}/\sigma_{\text{gross}}$ uses the gross-section nominal stress. The net-section factor is:

$$K_t = K_{tg} \left(1 - \frac{D}{W} \right) \quad (2.6)$$

Notch sensitivity and fatigue strength reduction factor:

$$K_f = 1 + q(K_t - 1), \quad 0 \leq q \leq 1 \quad (2.7)$$

where q is the notch sensitivity factor. For $q = 0$ (fully insensitive material), $K_f = 1$; for $q = 1$ (fully sensitive), $K_f = K_t$.

Problem 1.

Creager–Paris: maximum stress at a crack tip

Consider an infinite plate containing a central through-thickness crack of total length $2a = 3.80 \times 10^{-2}$ mm, with a rounded tip of radius $\rho = 1.90 \times 10^{-4}$ mm (Figure 2.1). The plate is subjected to a uniform remote tensile stress $\sigma_\infty = 140$ MPa.

The maximum elastic stress at the tip of a crack of half-length a with a finite root radius ρ , under a remote uniform stress σ_∞ , is given by the Creager–Paris solution:

$$\sigma_m = \sigma_\infty \left(1 + 2\sqrt{\frac{a}{\rho}} \right). \quad (2.8)$$

For sharp defects ($a \gg \rho$), this reduces to the approximate form:

$$\sigma_m \approx 2\sigma_\infty \sqrt{\frac{a}{\rho}}. \quad (2.9)$$

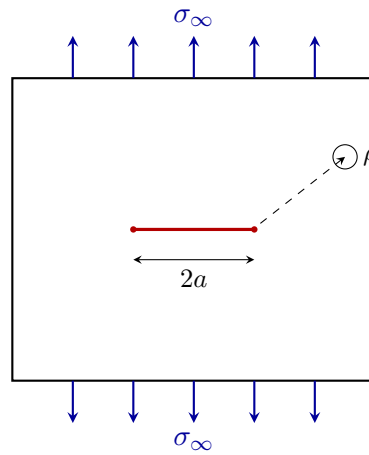


Figure 2.1: Infinite plate with a central crack under uniform remote tension.

- Using the Creager–Paris solution, compute the maximum elastic stress σ_m at the crack tip, and the corresponding stress concentration factor K_t .
- Repeat the calculation using the approximate form, valid for sharp defects ($a \gg \rho$). Compute the relative error with respect to the exact Creager–Paris result.
- Comment on the physical meaning of the result: what happens to σ_m as $\rho \rightarrow 0$?

Problem 2.

Theoretical fracture strength from the Creager–Paris solution

A semi-infinite plate of a brittle material contains an edge crack of length $a = 0.50$ mm with a tip radius $\rho = 5 \times 10^{-4}$ mm. Fracture occurs when the remote applied stress reaches $\sigma_\infty = 1035$ MPa.

- Using the Creager–Paris solution, compute the maximum stress at the crack tip at the instant of fracture. This value corresponds to the theoretical fracture strength of the material.
- Compare the computed value with the theoretical cohesive strength $\sigma_{th} \approx E/10$ for a typical ceramic ($E \approx 400$ GPa). Comment on the order of magnitude.

Problem 3.

Stress concentration factor for an internal defect

Consider a large component containing an internal defect of total length $2a = 3$ mm and a tip radius $\rho = 0.05$ mm, subjected to uniform remote tensile loading.

- Determine the stress concentration factor K_t for this defect using the Creager–Paris solution.
- Explain the physical significance of the computed K_t value: by how much is the stress amplified at the defect tip compared to the remotely applied stress?
- If the material has a yield stress $\sigma_e = 300$ MPa, estimate the remote stress level σ_∞ at which localised yielding first occurs at the defect tip (assuming linear-elastic behaviour elsewhere).

Problem 4.

K_t definitions: net section vs. gross section

The stress concentration factor can be defined using either the net-section nominal stress (K_t) or the gross-section nominal stress (K_{tg}).

- Which definition is the conventional one used in reference handbooks (*e.g.* Peterson’s *Stress Concentration Factors*)? Briefly justify why this convention is adopted.
- Write the general definition of K_t and explain the assumptions behind it (elastic behaviour, nominal stress definition).
- Show that for a strip of width W with a central hole of diameter D , the relationship $K_t = K_{tg}(1 - D/W)$ holds. Is K_{tg} always smaller than K_t ? Under what conditions do both definitions converge?

Problem 5.

Stress field around a circular hole in an infinite plate

An infinite sheet containing a circular hole of diameter $D = 2a$ is loaded in uniaxial tension by a remote stress S along the y -direction.

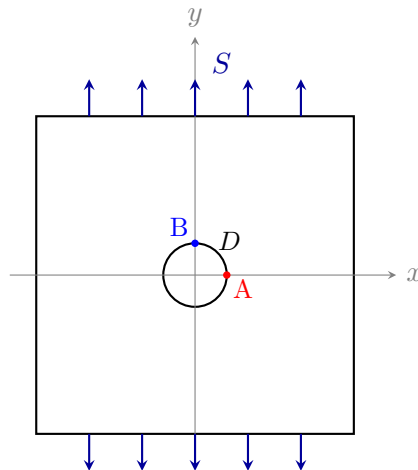


Figure 2.2: Infinite sheet with a circular hole under uniaxial tension.

The classical Kirsch solution gives the tangential stress along the edge of the hole ($r = a$) as

$$\sigma_\theta(S, \theta) = S(1 + 2 \cos 2\theta),$$

where θ is measured from point A, located on the positive x -axis. Therefore, point A corresponds to $\theta = 0^\circ$, and point B corresponds to $\theta = 90^\circ$.

Along the x -axis, starting from point A and moving away from the hole, the normal stress component is

$$\frac{\sigma_y(r)}{S} = 1 + \frac{1}{2} \left(\frac{a}{r}\right)^2 + \frac{3}{2} \left(\frac{a}{r}\right)^4, \quad r \geq a.$$

The corresponding stress gradient at the edge of the hole is

$$\left. \frac{d\sigma_y}{dr} \right|_{r=a} = -\frac{7S}{a}.$$

- Using the tangential stress expression, determine the value of K_t for a circular hole in an infinite plate.
- Compute the tangential stress σ_θ at the top of the hole, point B. Is this stress tensile or compressive?
- Describe qualitatively how the tangential stress σ_θ varies along the edge of the hole, from point A ($\theta = 0^\circ$) to point B ($\theta = 90^\circ$). At which angle does σ_θ become zero?
- Explain how the stress component σ_y decays along the x -axis away from point A. Use the expression for $\sigma_y(r)$ and the stress gradient at the hole edge to comment on the rate of decay.

Problem 6.

Circular hole in a finite-width strip

A strip of width W containing a central circular hole of diameter D is loaded in uniaxial tension.

- Using the Howland approximation (Eq. 2.5), sketch K_t (net-section) as a function of D/W for $0 \leq D/W \leq 0.6$. Identify the value of D/W at which K_t is a minimum and the corresponding minimum K_t value.
- Explain why K_t decreases for small D/W (approaching the infinite-plate limit) and then increases again as D/W grows (why does K_t eventually rise?).
- Numerical application:** A strip has the following dimensions: $W = 60$ mm, $D = 10$ mm, thickness $t = 5$ mm. The strip is loaded with $P = 50$ kN.
 - Compute K_{tg} and K_t for this geometry.
 - Determine the peak stress σ_{peak} at the edge of the hole.
- Does K_t depend on the applied load P ? Justify.

Problem 7.

Elliptical hole and the transition to a crack

An infinite plate is loaded in uniaxial tension (stress S) and contains an elliptical hole with semi-axes a (perpendicular to the load) and b (parallel to the load).

- (a) Write the expression for K_t of the elliptical hole (Inglis solution) and verify that for $a = b$ (circular hole) it recovers $K_t = 3$.
- (b) Rewrite K_t as a function of a and the tip radius $\rho = b^2/a$. Show that this is equivalent to the Creager–Paris formula.
- (c) Plot (or sketch) K_t as a function of a/b for $a/b \in [1, 20]$. Comment on what happens as the ellipse becomes increasingly elongated ($a/b \rightarrow \infty$, $\rho \rightarrow 0$): why does the concept of K_t become impractical, and what parameter replaces it?

Problem 8.

Notch sensitivity and K_f

The fatigue strength reduction factor K_f is typically smaller than the elastic stress concentration factor K_t , reflecting the fact that not all of the stress concentration translates into an equivalent reduction in fatigue strength.

- (a) Define the notch sensitivity factor q and explain the meaning of the limits $q = 0$ and $q = 1$.
- (b) Is the fact that $K_f < K_t$ favourable for the designer? Why does this difference arise from a physical standpoint (consider the stress gradient at the notch root and the microstructural scale of crack initiation)?
- (c) Compare the notch sensitivity of a low-strength steel with that of a high-strength steel. Which material is more notch-sensitive? Briefly explain why.
- (d) The prediction of the fatigue limit of a notched component is usually based on a comparison with the fatigue limit of unnotched material (the *similarity concept*). Discuss possible limitations of this approach.

Problem 9.

Surface finish, size effects and fatigue

Answer the following conceptual questions.

- (a) Size effects on fatigue can occur in notched components. What is the “relevant size” of the notch that governs this effect? Are the prediction methods for size effects on fatigue fundamentally rational or empirical?
- (b) Compare the significance of surface finish for: (i) the fatigue limit, (ii) high-cycle fatigue life, and (iii) low-cycle fatigue life. In which regime is surface finish most critical, and why?
- (c) Explain why a fully rational procedure to predict the complete S–N curve of a notched component does not currently exist, despite the availability of accurate K_t values.

Problem 10.

Stress concentration under biaxial loading

- (a) Consider an open hole in a large plate loaded under *pure shear*. What is the value of K_t in this case? Is the loading at the critical point on the hole edge uniaxial or biaxial?
- (b) A large sheet with a circular hole is subjected to biaxial tension with principal stresses S_1 (horizontal) and $S_2 = \beta S_1$ (vertical), where $\beta = 0.5$ is the biaxiality ratio. The stresses are applied fully in phase (synchronised).

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- i) Using the superposition principle, determine the tangential stress at the critical point (point A, where the stress is maximum) in terms of S_1 .
 - ii) Now consider the case where S_1 and S_2 are *not* synchronised: S_1 cycles between 0 and $S_1^{\max}$ while $S_2 = \beta S_1^{\max}$ is constant (static). Determine the stress range at point A.
- (c) Comment on why the synchronisation of the biaxial loading is important for fatigue assessment.

Answer:  [Open in Colab](#)

3 Problems 3

Formulary

Stress intensity factor (general form, Mode I):

$$K_I = Y \sigma \sqrt{\pi a} \quad (3.1)$$

where σ is the remote (nominal) stress, a is the crack half-length (for a central crack) or full length (for an edge crack), and $Y = Y(a/W)$ is a dimensionless geometry correction factor. For an infinite plate with a central crack, $Y = 1$.

Centre-cracked tension (CCT) panel of width W — Feddersen (secant) approximation:

$$Y\left(\frac{a}{W}\right) = \sqrt{\sec\left(\frac{\pi a}{W}\right)} \quad (3.2)$$

Single-edge-notched three-point bending (SEN-3PB) specimen ($S = 4W$, ASTM E399):

$$K_I = \frac{PS}{BW^{3/2}} f\left(\frac{a}{W}\right), \quad (3.3)$$

$$f\left(\frac{a}{W}\right) = \frac{3\sqrt{a/W}}{2(1+2a/W)(1-a/W)^{3/2}} \left[1.99 - \frac{a}{W} \left(1 - \frac{a}{W}\right) \left(2.15 - 3.93 \frac{a}{W} + 2.7 \left(\frac{a}{W}\right)^2 \right) \right] \quad (3.4)$$

Compact tension (CT) specimen (ASTM E399):

$$K_I = \frac{P}{B\sqrt{W}} f\left(\frac{a}{W}\right), \quad (3.5)$$

$$f\left(\frac{a}{W}\right) = \frac{(2+a/W)}{(1-a/W)^{3/2}} \left[0.886 + 4.64 \frac{a}{W} - 13.32 \left(\frac{a}{W}\right)^2 + 14.72 \left(\frac{a}{W}\right)^3 - 5.6 \left(\frac{a}{W}\right)^4 \right] \quad (3.6)$$

Fracture condition:

$$K_I = K_{IC} \quad \implies \quad \sigma_f = \frac{K_{IC}}{Y\sqrt{\pi a}} \quad (3.7)$$

Critical crack size:

$$a_c = \frac{1}{\pi} \left(\frac{K_{IC}}{Y\sigma} \right)^2 \quad (3.8)$$

ASTM E399 plane-strain validity condition:

$$B, (W-a) \geq 2.5 \left(\frac{K_Q}{\sigma_e} \right)^2 \quad (3.9)$$

where K_Q is the provisional fracture toughness and σ_e is the 0.2% offset yield stress.

Residual strength diagram (Feddersen approach): the failure stress $\sigma_f(a)$ is the lower envelope of: (i) the LEFM curve, $\sigma_f = K_{IC}/(Y\sqrt{\pi a})$, and (ii) the net-section plastic collapse line, $\sigma_f = \sigma_e (1 - 2a/W)$, joined by a tangent line.

Problem 1.

The stress intensity factor: definitions and concepts

The stress intensity factor K is defined by the equation

$$K = \beta S \sqrt{\pi a}.$$

- Explain the meaning of each symbol in the equation: β , S , a .
- What is the unit of K in the International System of Units (SI)? What unit is commonly used in UK/US practice? Provide the conversion factor between the two.
- How would you define the stress intensity factor in words? What does it quantify physically?
- Which basic assumptions underlie the definition of K (material behaviour, crack geometry, loading)?
- K and K_t are fundamentally different concepts. Explain why. However, the parameters β (in the K definition) and K_t serve a similar purpose. Which purpose is that?

Problem 2.

Fracture assessment of a steel component

A specimen of AISI 4340 steel has a plane-strain fracture toughness $K_{IC} = 54.8 \text{ MPa}\sqrt{\text{m}}$ and is subjected to a remote tensile stress of $\sigma = 1030 \text{ MPa}$. The specimen contains an edge crack of length $a = 0.50 \text{ mm}$. Assume a geometry correction factor $Y = 1$ (semi-infinite plate).

- Compute the applied stress intensity factor K_I .
- Determine the fracture stress σ_f for this crack size.
- Will the specimen fracture under the applied stress? Justify by comparing K_I with K_{IC} , or equivalently, σ with σ_f .
- Compute the critical crack size a_c for the applied stress $\sigma = 1030 \text{ MPa}$. Interpret the result: if $a < a_c$, no fracture occurs.

Problem 3.

Geometry correction factor for an aircraft component

An aircraft component is fabricated from a high-strength aluminium alloy with a plane-strain fracture toughness $K_{IC} = 35 \text{ MPa}\sqrt{\text{m}}$. The component was designed so that, in the presence of an internal crack of total length $2a = 2.00 \text{ mm}$, fracture only occurs for a tensile stress of 250 MPa .

- Determine the geometry correction factor Y used in the design (recall that a in Eq. 3.1 is the half-crack length for a central crack).
- A subsequent inspection reveals an internal crack of total length $2a = 1.00 \text{ mm}$. Using the Y factor computed in (a), determine whether fracture will occur if a stress of 325 MPa is applied.
- Discuss the physical meaning of $Y > 1$: what kind of geometric or loading configuration might lead to such an amplification of the stress intensity factor?

Problem 4.**Detectability of a critical crack**

A wide structural component (plate with large dimensions) is to be fabricated from a high-strength steel with $K_{IC} = 77 \text{ MPa}\sqrt{\text{m}}$ and $\sigma_e = 1400 \text{ MPa}$. The minimum resolution of the available non-destructive inspection (NDI) equipment is 4 mm (i.e. any defect with total length $\geq 4 \text{ mm}$ can be detected). The design stress is $\sigma = \sigma_e/2 = 700 \text{ MPa}$. Assume $Y = 1$ and consider a central crack ($2a$).

- (a) Calculate the critical half-crack length a_c under the design stress.
- (b) Determine whether the critical crack ($2a_c$) would be detectable by the NDI equipment.
- (c) If instead the crack were an edge crack (length a , still with $Y = 1$), would the critical crack still be detectable? Discuss the design implications if the answer is negative.
- (d) What is the maximum design stress that ensures the critical *edge* crack is detectable (i.e. $a_c \geq 4 \text{ mm}$)?

Problem 5.**ASTM E399 fracture toughness test: SEN-3PB specimen**

A fracture test was conducted on a single-edge-notched three-point bending (SEN-3PB) specimen made from a high-strength low-alloy steel, following the ASTM E399 standard. The specimen has the following dimensions and properties:

Thickness	$B = 25 \text{ mm}$
Width (height)	$W = 50 \text{ mm}$
Support span	$S = 4W = 200 \text{ mm}$
Crack length (from fracture surface)	$a = 23.75 \text{ mm}$
Yield stress (0.2%)	$\sigma_e = 1640 \text{ MPa}$

The load–displacement record (Figure 3.1) is of ASTM E399 Type III (linear to failure), with $P_Q = P_{\max} = 27.2 \text{ kN}$. Therefore, $P_{\max}/P_Q = 1.00 < 1.10$, satisfying the first validity criterion.

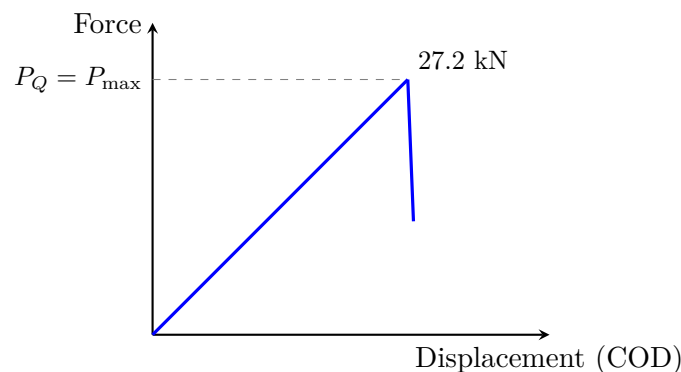


Figure 3.1: Schematic load–displacement record (Type III) for the SEN-3PB test.

- (a) Compute the ratio a/W and evaluate the shape function $f(a/W)$ using Eq. (3.4).
- (b) Calculate the provisional fracture toughness K_Q .
- (c) Convert K_Q from $\text{N} \cdot \text{mm}^{-3/2}$ to $\text{MPa}\sqrt{\text{m}}$. State the conversion factor.

- (d) Apply the ASTM E399 ligament validity check (Eq. 3.9): compute $2.5(K_Q/\sigma_e)^2$ and compare with $(W - a)$. Is K_Q a valid K_{IC} value?
- (e) If the validity check in (d) had failed, what practical modification to the specimen would you recommend to obtain a valid K_{IC} ?

Problem 6.

ASTM E399 fracture toughness test: CT specimen

A fracture test was conducted on a compact tension (CT) specimen of forged steel, following the ASTM E399 standard. The specimen has the following dimensions and properties:

Thickness	$B = 50$ mm
Width	$W = 100$ mm
Crack length	$a = 52$ mm
Yield stress	$\sigma_e = 1050$ MPa

The load–displacement record is of ASTM E399 Type I, with $P_Q = 241$ kN and $P_{\max} = 261$ kN.

- (a) Verify the first validity criterion: is $P_{\max}/P_Q \leq 1.10$?
- (b) Compute a/W and evaluate the shape function $f(a/W)$ using Eq. (3.6).
- (c) Calculate the provisional fracture toughness K_Q (in $\text{MPa}\sqrt{\text{m}}$).
- (d) Apply the ASTM E399 ligament validity check. Is the K_Q value valid as K_{IC} ?
- (e) If the test is not valid, compute the maximum K_Q value that would pass the validity check with the present specimen dimensions. Comment on what this implies about the material's ductility.

Problem 7.

Centre-cracked panel: SIF and fracture stress

A thick panel of width $W = 200$ mm, made from a high-strength aluminium alloy, contains a central through-thickness crack of total length $2a = 80$ mm. The panel fractures under a remote tensile stress of 100 MPa.

The Feddersen (secant) correction for a centre-cracked panel of finite width is given by Eq. (3.2):

$$Y(a/W) = \sqrt{\sec\left(\frac{\pi a}{W}\right)}.$$

- (a) Compute the geometry correction factor $Y(a/W)$ for this panel.
- (b) Determine the fracture toughness K_{IC} of the aluminium alloy.
- (c) Calculate the fracture stress for an *infinite* plate (i.e. $Y = 1$) with the same crack size $2a = 80$ mm. Compare with the finite-width result and comment.
- (d) Compute the fracture stress for a narrower panel of width $W = 120$ mm, with the same crack size. Compare with the result in (a).

Problem 8.**SEN-3PB: failure load prediction**

A standardised SEN-3PB specimen is to be tested in the same high-strength aluminium alloy as Exercise 3.7. The specimen dimensions are: $B = 50$ mm, $W = 100$ mm, $S = 2W = 200$ mm, and a fatigue pre-crack of length $a = 53$ mm was generated. Assume linear-elastic conditions.

- (a) Using the K_{IC} value determined in Exercise 3.7 and the SEN-3PB shape function (Eq. 3.4), compute the predicted failure load P_f .
- (b) If the aluminium alloy has a yield stress $\sigma_e = 450$ MPa, verify whether the ASTM E399 plane-strain validity conditions are satisfied for this specimen geometry and material.

Problem 9.**ASTM E399 validity: minimum yield stress**

Consider the SEN-3PB specimen of Exercise 3.8 ($B = 50$ mm, $W = 100$ mm, $a = 53$ mm) and assume the same aluminium alloy with K_{IC} from Exercise 3.7.

- (a) If the yield stress of the alloy is $\sigma_e = 450$ MPa, verify whether the ASTM E399 plane-strain condition (Eq. 3.9) is met for both the thickness B and the ligament ($W - a$).
- (b) Determine the minimum yield stress $\sigma_{e,\min}$ that the alloy must possess (for the same K_{IC}) in order for the present specimen dimensions to provide a valid K_{IC} measurement. Which dimension (thickness or ligament) is the controlling constraint?

Problem 10.**Residual strength diagram**

Consider again the high-strength aluminium panel of Exercise 3.7 ($W = 200$ mm, K_{IC} as computed). The alloy has a yield stress $\sigma_e = 450$ MPa.

Construct the residual strength diagram $\sigma_f(a)$ for $0 \leq a \leq W/2$ by plotting:

- (a) The LEFM fracture curve: $\sigma_f(a) = K_{IC} / [Y(a/W) \sqrt{\pi a}]$, where $Y(a/W)$ is the Feddersen correction.
- (b) The net-section plastic collapse line: $\sigma_f(a) = \sigma_e (1 - 2a/W)$.
- (c) The Feddersen construction: connect the two curves with a smooth tangent, so that the residual strength is the lower envelope of LEFM and plastic collapse, joined by the tangent.
- (d) Identify the transition crack length a^* at which the failure mode changes from plastic collapse to brittle fracture. Comment on its engineering significance.

Problem 11.**Conceptual questions on K and crack-tip behaviour**

Answer the following conceptual questions. Brief but precise answers are expected.

- (a) Is it possible for the geometry correction factor β (or Y) to have a constant value, independent of the crack length? Give an example.
- (b) The K -field predicts an infinite stress at the crack tip ($r \rightarrow 0$). In practice, a small plastic zone forms instead.
 - a) Is the K -field equation valid inside the plastic zone?

- b) Explain why K can still be a useful parameter, despite the existence of a small plastic zone at the crack tip (*small-scale yielding* argument).
- (c) Why is the crack tip of a through-thickness crack in a *thick* plate approximately in plane strain at low stress levels, while the same does not hold for a *thin* sheet?
- (d) What is the meaning of the *crack driving force*? How does it relate to the energy release rate G and the stress intensity factor K ?

Answer:  [Open in Colab](#)

4 Problems 4

Formulary

Paris–Erdogan law:

$$\frac{da}{dN} = C (\Delta K)^m \quad (4.1)$$

where C and m are material constants, and $\Delta K = K_{\max} - K_{\min} = Y \Delta \sigma \sqrt{\pi a}$.

Stress intensity factor range for a CCT panel (Feddersen):

$$\Delta K = Y(a/W) \Delta \sigma \sqrt{\pi a}, \quad Y(a/W) = \left[1 - 0.1 \left(\frac{a}{W} \right)^2 + 0.96 \left(\frac{a}{W} \right)^4 \right] \sqrt{\sec \left(\frac{\pi a}{W} \right)} \quad (4.2)$$

Number of cycles by integration:

$$N_f = \int_{a_0}^{a_c} \frac{da}{C [\Delta K(a)]^m} \quad (4.3)$$

Fracture condition ($K_{\max} = K_{IC}$):

$$a_c : \quad Y(a_c/W) \frac{\Delta \sigma}{1 - R} \sqrt{\pi a_c} = K_{IC} \quad (4.4)$$

Relationship between Paris law and S–N curve: For infinite-plate geometry ($Y = 1$, constant), integrating Paris’s law yields:

$$N_f \propto (\Delta \sigma)^{-m} \quad \Longleftrightarrow \quad (\Delta \sigma)^m N_f = \text{const.} \quad (4.5)$$

recovering the S–N power-law form with slope exponent equal to the Paris exponent m .

Problem 1.

Conceptual questions on fatigue crack growth

- Which conditions must be satisfied for the application of the similitude concept to predict fatigue crack growth in a component under constant-amplitude loading?
- Describe the three regions of a typical da/dN vs. ΔK plot (log–log). What physical mechanism dominates in each region?
- What is the threshold stress intensity factor range ΔK_{th} ? What is the “small crack” problem associated with ΔK_{th} ?
- How should constant- ΔK tests be carried out? What is one advantage and one disadvantage of this type of test?

Problem 2.

Paris law: units and sensitivity

The Paris relation for SAE 4340 steel can be written as:

$$\frac{da}{dN} = 17.9 \times 10^{-6} (\Delta K)^{2.675}$$

with da/dN in mm/kcycle and ΔK in $\text{MPa}\sqrt{\text{m}}$.

- (a) Rewrite this relation using the units: da/dN in inch/cycle and ΔK in $\text{ksi}\sqrt{\text{in}}$.
- (b) Are the constants C and m dependent on the unit system? Explain.
- (c) A component is loaded under CA loading. If the stress level is increased by 30%, by what factor does the crack growth rate increase? Use the Paris exponent $m = 2.675$.

Problem 3.

Crack closure and R -ratio effects

- (a) Explain the concept of crack closure (Elber). What is ΔK_{eff} and how does it differ from ΔK ?
- (b) Is the $da/dN = f(\Delta K)$ relation affected by the stress ratio R ? Why?
- (c) Is crack growth affected by the material thickness? Explain in terms of plane stress vs. plane strain at the crack tip.
- (d) Under what circumstances can the R -dependence be eliminated by using ΔK_{eff} ?

Problem 4.

Paris law integration for a CCT panel

Consider the rectangular plate of width $W = 100$ mm shown in Figure 4.1, containing a central through-thickness crack of initial total length $2a_0 = 1$ mm. The plate is made of a carbon steel (0.13%C) with yield stress $\sigma_e = 326$ MPa and fracture toughness $K_{\text{IC}} = 50 \text{ MPa}\sqrt{\text{m}}$. The plate is subjected to constant-amplitude fatigue loading with:

$$\Delta\sigma = 200 \text{ MPa}, \quad R = 0.25.$$

The Paris law for this material is:

$$\frac{da}{dN} = 1 \times 10^{-15} (\Delta K)^3$$

with da/dN in mm/cycle and ΔK in $\text{N}\cdot\text{mm}^{-3/2}$.

The stress intensity factor range is given by Eq. (4.2).

- (a) Compute the maximum stress σ_{max} from $\Delta\sigma$ and R .
- (b) Determine the critical crack half-length a_c by solving $K_{\text{max}}(a_c) = K_{\text{IC}}$ numerically (or iteratively).
- (c) Implement the cycle-by-cycle (or incremental Δa) numerical integration of the Paris law from a_0 to a_c using Python. Use a fixed crack increment Δa (e.g. 0.1 mm) and the forward Euler scheme:

$$\Delta N_i = \frac{\Delta a}{C [\Delta K(a_i)]^m}, \quad N_{i+1} = N_i + \Delta N_i, \quad a_{i+1} = a_i + \Delta a.$$

- (d) Plot a vs. N and da/dN vs. ΔK (log-log) from your numerical integration.
- (e) Report the total fatigue life N_f (number of cycles from a_0 to a_c).
- (f) Verify the convergence of your solution by reducing Δa by a factor of 2 and comparing N_f .

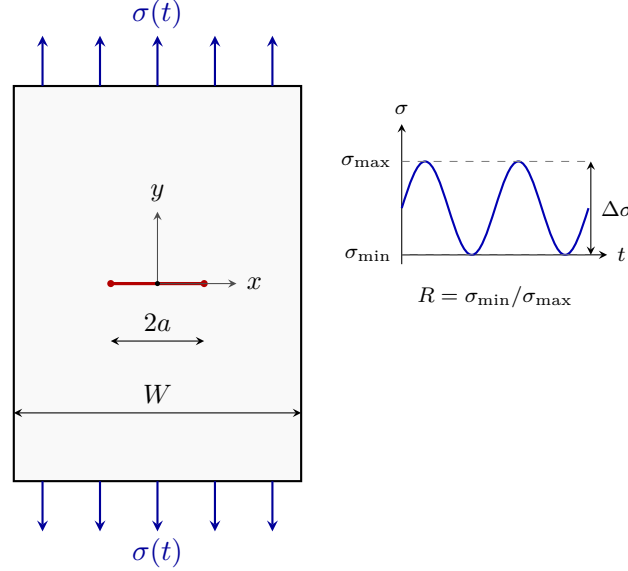


Figure 4.1: Central-cracked tension (CCT) panel of width W with a through-thickness central crack of total length $2a$, subjected to remote constant-amplitude cyclic stress of range $\Delta\sigma$ and stress ratio $R = \sigma_{\min}/\sigma_{\max}$. The crack length is exaggerated for clarity ($2a_0/W = 1/100$ in the problem).

Problem 5.

Paris law recovers the S–N power law

Consider an infinite plate ($Y = 1$) with a central crack under CA loading. The Paris law is $da/dN = C (\Delta K)^m$.

- (a) Show that by integrating the Paris law from a_0 to a_c (with $Y = 1$ and $\Delta K = \Delta\sigma\sqrt{\pi a}$), the fatigue life takes the form:

$$N_f = \frac{1}{C (\Delta\sigma)^m \pi^{m/2}} \int_{a_0}^{a_c} a^{-m/2} da.$$

- (b) For $m > 2$ and $a_c \gg a_0$, show that $N_f \approx \text{const.} \times (\Delta\sigma)^{-m}$, which is equivalent to the S–N power-law relation $(\Delta\sigma)^m N_f = \text{const.}$
- (c) Discuss the physical interpretation: the Paris exponent m directly equals the S–N slope exponent when crack initiation is negligible.

Problem 6.

Incremental crack growth with width correction

A sheet of width $W = 160$ mm contains a central hole of diameter $D = 8$ mm with two symmetric edge cracks of initial length $a_0 = 4$ mm (effective initial crack: $a_{\text{eff},0} = a_0 + D/2 = 8$ mm). The sheet is loaded under CA with $R = 0$, $S_{\max} = 80$ MPa. Crack growth data for the material are available from the Schijve textbook (Figure E8.11b).

- (a) Predict the crack growth from $a = 4$ mm to $a = 36$ mm using incremental steps of $\Delta a = 4$ mm. Apply the Feddersen width correction at each step. Tabulate: a_i , $a_{\text{eff},i}$, $Y(a_{\text{eff},i}/W)$, ΔK_i , $(da/dN)_i$, ΔN_i , $\sum N$.
- (b) Plot a vs. N and comment on the acceleration of crack growth as $a \rightarrow W/2$.
- (c) Extract the Paris constants C and m from the da/dN vs. ΔK data by log-log regression.
- (d) Compare the $Y(a/W)$ correction obtained with the Feddersen formula against the simpler approximation $Y = 1$. How much does the fatigue life change?

Answer:  [Open in Colab](#)

5 Problems 5

Formulary

Soderberg relation:

$$\frac{\sigma_a}{\sigma_{f0}} + \frac{\sigma_m}{\sigma_e} = 1$$

Goodman relation:

$$\frac{\sigma_a}{\sigma_{f0}} + \frac{\sigma_m}{\sigma_u} = 1$$

Gerber relation:

$$\frac{\sigma_a}{\sigma_{f0}} + \left(\frac{\sigma_m}{\sigma_u} \right)^2 = 1$$

where σ_{f0} is the fatigue limit for $R = -1$, σ_e the yield stress, and σ_u the ultimate tensile strength.

Modified Basquin (mean stress correction):

$$\sigma_a = (\sigma'_f - \sigma_m) (2N_f)^b$$

Miner's linear damage accumulation rule:

$$D = \sum_{i=1}^k \frac{n_i}{N_{f,i}} \leq 1$$

where n_i is the number of cycles applied at block i and $N_{f,i}$ is the fatigue life under that constant-amplitude loading alone.

Problem 1.

Soderberg diagram for a forged steel

A forged steel has: $\sigma_e = 386$ MPa, $\sigma_u = 620$ MPa, fatigue limit (unnotched, $R = -1$) $\sigma_{f0} = 296$ MPa.

- Construct the Soderberg diagram (σ_a vs. σ_m).
- Determine the fatigue limit (amplitude) for a loading with $\sigma_m = \sigma_a/2$.
- Determine the fatigue limit for a pulsating loading ($R = 0$, *i.e.* $\sigma_{\min} = 0$).
- On the same diagram, construct the Goodman and Gerber lines. Compare the predictions for parts (b) and (c).

Problem 2.

Effect of mean stress on specimen ranking

Three identical fatigue specimens (A, B, C) of a non-ferrous alloy are tested under the conditions shown:

Specimen	$\sigma_{\max}$ (MPa)	$\sigma_{\min}$ (MPa)
A	+450	−150
B	+300	−300
C	+500	−200

- Compute σ_m and σ_a for each specimen.
- Rank the specimens from longest to shortest fatigue life. Justify using a Soderberg-type argument.
- Sketch two S–N curves (for two different σ_m values) and locate the three test points.

Problem 3.

S–N curve with Eurocode 3 and effect of stress reduction

Using the EC3 S–N curve from Week 1 ($m = 3$, $\Delta\sigma_D = 160$ MPa):

- A component operates at $\Delta\sigma = 300$ MPa. Compute N_f .
- If the stress range is reduced by 10% (to 270 MPa), compute the new N_f and the percentage increase in life.
- Generalise: plot the percentage increase in N_f as a function of the percentage reduction in $\Delta\sigma$ (from 0 to 20%), starting from 300 MPa. Comment on the non-linear sensitivity.

Problem 4.

Miner’s rule: AISI 1045 steel

An AISI 1045 steel component is subjected to the following block loading spectrum ($\sigma'_f = 1198.6$ MPa, $b = -0.09$; modified Basquin with mean stress correction):

$\sigma_{\max}$ (MPa)	$\sigma_{\min}$ (MPa)	σ_m (MPa)	n (cycles)
+450	−150	+150	300 000
+300	−300	0	150 000
+500	−200	+150	30 000

- For each block, compute σ_a and $N_{f,i}$ using the modified Basquin relation.
- Compute the cumulative Miner damage $D = \sum n_i/N_{f,i}$.
- Does the component survive the applied spectrum? If so, how many more identical spectra could be applied before failure?

Problem 5.

Miner’s rule: shortcomings

- The Miner rule has three well-known shortcomings. State them.

-
- (b) Can these shortcomings lead to $\sum n/N$ values both smaller and larger than 1? Explain.
 - (c) The Miner rule ignores damage from cycles with amplitude below the fatigue limit. How can this be accounted for approximately? Is this correction fully rational?
 - (d) Is the sequence of cyclic loads significant for fatigue under variable-amplitude loading? Give an example.

Problem 6.

Estimating the S–N curve from σ_u

For a forged steel with $\sigma_u = 1000$ MPa:

- (a) Estimate the fatigue limit ($R = -1$) using $\sigma_{f0} \approx 0.5 \sigma_u$.
- (b) Estimate the fatigue strength at $N_f = 10^3$ cycles using $\sigma_a(10^3) \approx 0.9 \sigma_u$.
- (c) Construct the estimated S–N curve (Basquin form) passing through these two points.
- (d) Predict the fatigue strength at $N_f = 10^4$ cycles.

Problem 7.

Residual stresses

- (a) Give a definition of a residual stress distribution. What is characteristic about the equilibrium of a residual stress field?
- (b) What is the effect of residual stresses on fatigue? Distinguish between compressive and tensile residual stresses.
- (c) Residual stresses can be introduced intentionally (*e.g.* shot peening, plastic hole expansion) or unintentionally (production, heat treatment). Give one example of each.

Answer:  [Open in Colab](#)

6 Problems 6

Problem 1.

Effect of R -ratio on fatigue life

Using the CCT panel from Exercise 4.4 (Week 4), with $\Delta\sigma = 200$ MPa, $W = 100$ mm, $2a_0 = 1$ mm:

- (a) Compute the fatigue life N_f for $R = 0, 0.1, 0.25, 0.5$. Note that $\sigma_{\max} = \Delta\sigma/(1 - R)$ changes with R , which affects a_c .
- (b) Plot all four $a(N)$ curves on the same graph.
- (c) Discuss the trend: why does increasing R (for the same $\Delta\sigma$) reduce N_f ?

Problem 2.

Effect of initial crack size

Using the same panel and $R = 0.25$:

- (a) Compute N_f for $2a_0 = 0.5, 1.0, 2.0, 5.0$ mm.
- (b) Plot N_f vs. a_0 on a semi-log scale.
- (c) Discuss the sensitivity of fatigue life to the initial defect size and its implications for non-destructive inspection.

Problem 3.

SEN-3PB specimen: ΔP - N_f curve

Consider a SEN-3PB specimen with $B = 10$ mm, $W = 30$ mm, $S = 150$ mm, initial crack $a_0 = 0.25$ mm, and $K_{IC} = 50$ MPa $\sqrt{\text{m}}$. The material follows the same Paris law as Exercise 4.4. $R = 0.3$.

- (a) For applied load ranges $\Delta P = 0.5, 1.0, 1.5, 2.0, 2.5$ kN, compute N_f by Paris integration using the SEN-3PB shape function.
- (b) Plot the ΔP - N_f curve on a log-log scale. Compare with a straight line (power law): does the SEN-3PB geometry produce a linear ΔP - N_f relationship?

Problem 4.

Crack growth predictions: accuracy considerations

- (a) Consider a given value of $\Delta a = 4$ mm. Discuss how the choice of Δa affects the accuracy of the predicted N_f . Is the forward Euler scheme conservative or non-conservative?
- (b) What additional factors should be considered for accurate crack growth predictions in real structures (e.g. crack closure, retardation, mixed-mode loading, environmental effects)?

Problem 5.

Variable-amplitude crack growth: interaction effects

- (a) What is meant by “non-interaction” during fatigue crack growth under VA loading?
- (b) What is the effect of a single overload cycle on subsequent CA crack growth? And of an underload?
- (c) What is “delayed retardation” after an overload?
- (d) Discuss the effect of: (i) material thickness, (ii) increasing yield stress, and (iii) a block of overloads (vs. a single OL) on crack growth retardation.

Answer:  [Open in Colab](#)

7 Problems 7

Problem 1.

Fatigue of lugs and bolted joints

- (a) Why is the fatigue limit of a lug very low compared to an open-hole specimen? Why is the size effect different?
- (b) For fatigue of non-symmetric joints (e.g. a lap joint), an additional fatigue aspect must be considered. What is it, and why is it unfavourable?
- (c) In a riveted or bolted lap joint, stress concentrations arise for different reasons. List them and suggest how the fatigue strength of the joint can be improved.
- (d) In a bolted joint (bolts in shear), the clamping force can be increased by tightening the nut. What is the effect on: (i) load transmission and (ii) fatigue life?
- (e) Explain why pre-tensioning a bolt (loaded in tension) is beneficial for fatigue.

Problem 2.

Fretting fatigue

- (a) Is fretting primarily a matter of surface damage, corrosion, or both?
- (b) What is the effect of fretting corrosion on an S-N curve? What can be said about the clamping pressure between mating surfaces?
- (c) What is the remedy against fretting corrosion?
- (d) Is fretting significant for lugs and for riveted/bolted joints? If a bolt connects two parts but does not transmit any load, is fretting still possible?

Problem 3.

Corrosion fatigue

- (a) Which periods of the fatigue life (initiation, growth) are affected by corrosion in a liquid environment?
- (b) Why is the effect of corrosion on the fatigue limit so large?
- (c) Is corrosion fatigue dependent on the loading frequency? On the wave shape? Explain.
- (d) For aluminium alloys, which gaseous component contributes to corrosion fatigue? Why can a low temperature be favourable?
- (e) What is the best general method to combat corrosion fatigue?

Problem 4.**Welded cruciform joint: S–N curve from Paris integration**

Consider a load-carrying cruciform welded joint with the following data:

Plate thickness	$T_p = 50$ mm
Weld leg	$H = 15$ mm
Root defect	$2a = 5$ mm
Weld angle	$\theta = 45^\circ$
Fracture toughness	$K_{IC} = 45$ MPa $\sqrt{\text{m}}$
R -ratio	$R = 0.25$
Paris law	$da/dN = 1 \times 10^{-15} (\Delta K)^3$ (mm/cycle, N $\cdot$ mm $^{-3/2}$)

The stress intensity factor for the root crack is:

$$K = \sigma \sqrt{\pi a} \left[A_1 + A_2 \left(\frac{a}{W} \right) \right] \sqrt{\sec \left(\frac{\pi a}{2W} \right)}$$

where $W = H + T_p$ and A_1, A_2 are polynomial functions of H/T_p (given in the problem sheet).

- (a) Compute W, A_1, A_2 for $H/T_p = 0.3$.
- (b) By Paris integration, determine the S–N curve of the joint: compute N_f for $\Delta\sigma = 100, 150, 200, 250, 300$ MPa.
- (c) Plot the computed S–N curve on a log–log scale. Compare with the IIW FAT 36 class (if applicable).

Problem 5.**Welded joints: concepts**

- (a) Fatigue of welded structures is fundamentally different from fatigue of non-welded structures. Mention two prominent differences.
- (b) When designing against fatigue of welded joints, which aspects are most relevant?
- (c) A connection between two parts of a dynamically loaded steel structure can be made as a welded joint or a bolted joint. How should this problem be handled from a fatigue perspective?

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8 Problems 8

Formulary

Given experimental data points (a_i, N_i) , the crack growth rate at the midpoint is:

$$\left(\frac{da}{dN}\right)_i = \frac{a_{i+1} - a_i}{N_{i+1} - N_i}, \quad a_{\text{mid},i} = \frac{a_{i+1} + a_i}{2}$$

The corresponding ΔK is evaluated at $a_{\text{mid},i}$ using the appropriate geometry factor.

Problem 1.

Critical review of Paris integration methodology

- (a) What are the key sources of error in a Paris law integration for predicting fatigue life?
- (b) Discuss the sensitivity of N_f to: (i) the Paris constants C and m , (ii) the initial crack size a_0 , and (iii) the geometry correction factor $Y(a/W)$.
- (c) How should the increment Δa be chosen to balance accuracy and computational cost? What convergence criterion would you recommend?

Problem 2.

Paris constants: units and conversion

- (a) The Paris relation $da/dN = C(\Delta K)^m$ has constants that depend on the unit system. If C and m are determined with da/dN in mm/cycle and ΔK in MPa $\sqrt{\text{m}}$, derive the conversion to: da/dN in mm/cycle and ΔK in N $\cdot\text{mm}^{-3/2}$.
- (b) Does the exponent m change with the unit system? Why or why not?

Problem 3.

Extracting Paris constants from experimental a - N data

Using the experimental a vs. N data recorded by your group in Week 9:

- (a) Apply the ASTM E647 secant method to compute da/dN and ΔK at each midpoint.
- (b) Plot da/dN vs. ΔK on a log-log scale.
- (c) Perform a linear regression on the log-log data to extract C and m . Report R^2 .
- (d) Compare your extracted C and m with literature values for the tested alloy. Discuss sources of discrepancy.

Problem 4.

Crack closure and R -ratio effects

- (a) From your experimental data, is there evidence of crack closure (e.g. load–displacement non-linearity at low loads)?
- (b) If your test was run at a single R -ratio, discuss how you would expect the Paris curve to shift for a different R -ratio.
- (c) Would the thickness of your specimen (plane stress vs. plane strain) affect the measured da/dN vs. ΔK data? Explain.

Answer:  [Open in Colab](#)

9 Problems 9

Problem 1.

Sources and characteristics of fatigue scatter

- (a) List the main sources of scatter in fatigue test results.
- (b) Compare the scatter typically observed in: (i) the crack initiation period and (ii) the crack growth period. In which period is scatter usually larger?
- (c) Is the scatter of the fatigue limit larger or smaller than the scatter of fatigue lives at high stress amplitudes? Why?
- (d) At a given stress amplitude, should the statistical distribution of fatigue lives be described by a Gaussian (normal), log-normal, or Weibull distribution? Justify.
- (e) What is the practical meaning of a p -S-N curve? How is it constructed?
- (f) How many specimens are needed to determine the fatigue limit with reasonable statistical confidence? Which test strategies exist (staircase method, etc.)?

Problem 2.

S-N data fitting: brass

Rotating-bending fatigue tests on a brass alloy yielded the following results:

σ_a (MPa)	N_f (cycles)
170	3.7×10^4
148	1.0×10^5
130	3.0×10^5
114	1.0×10^6
92	1.0×10^7
80	1.0×10^8
74	1.0×10^9

- (a) Plot the S-N data on a log-log scale.
- (b) Fit a power-law curve $\sigma_a = A N_f^b$ using log-log regression. Report A , b and R^2 .
- (c) Does this material exhibit a clear fatigue limit? Compare with the behaviour of a typical steel (BCC vs. FCC crystal structure).
- (d) Use the fitted curve to estimate: (i) N_f for $\sigma_a = 120$ MPa, and (ii) σ_a for $N_f = 4 \times 10^6$.

Problem 3.**S–N data fitting: steel with fatigue limit**

Rotating-bending fatigue tests on a steel alloy yielded:

σ_a (MPa)	N_f (cycles)
470	1×10^4
440	3×10^4
390	1×10^5
350	3×10^5
310	1×10^6
290	3×10^6
290	1×10^7
290	1×10^8

- (a) Plot the S–N data. Identify the fatigue limit.
- (b) Fit a power law to the finite-life region only (exclude the plateau points). Report A , b and R^2 .
- (c) This steel is used in an automotive shaft rotating at 600 rpm. Compute the expected service life (in hours) for $\sigma_a = 380$ MPa.
- (d) What is the maximum stress amplitude for the shaft to survive 1 year of continuous operation?

Problem 4.**Weibull distribution and POD concepts**

- (a) The two-parameter Weibull distribution is often used to model fatigue life scatter. Write its CDF and explain the meaning of the shape parameter β and scale parameter η .
- (b) Given 10 fatigue life results (in 10^3 cycles): 45, 52, 58, 63, 71, 78, 89, 105, 132, 190, estimate the Weibull parameters using a linearised plot (Weibull probability paper).
- (c) Define the Probability of Detection (POD) for an NDI technique. Why is the POD curve important for establishing inspection intervals in damage-tolerant design?
- (d) Explain the concept of the $a_{90/95}$ crack size in the context of POD.

Answer:  [Open in Colab](#)

10 Problems 10

Problem 1.

Identifying failure modes from fracture surfaces

- (a) Describe the macroscopic features of a fatigue fracture surface. How do you distinguish fatigue from: (i) static overload, (ii) stress corrosion cracking?
- (b) On a fatigue fracture surface, identify and explain: beach marks (arrest lines), ratchet marks, striations, and the final fast-fracture zone. What information does each feature convey about the loading history?
- (c) A failed component shows a smooth, flat fracture surface with concentric arc-shaped marks converging to a single point at the surface. What is the most likely failure mode? Where did the crack initiate?

Problem 2.

Fractography of three specimens

Three specimens were tested and their fracture surfaces examined. For each, determine: (a) the fracture type (ductile, brittle, fatigue), (b) the loading type (tension, bending, torsion), and (c) the probable failure origin.

- (a) A cylindrical shaft showing a smooth, flat fracture surface covering approximately 80% of the cross-section, with a small, rough region (shear lip) at the edge. Beach marks are visible, radiating from a surface scratch.
- (b) A rectangular plate showing a cup-and-cone fracture with extensive necking. The fracture surface is rough and dimpled.
- (c) A cylindrical component with a fracture surface at 45° to the axis, showing a smooth, helical pattern with no beach marks.

Problem 3.

de Havilland Comet case study

- (a) Describe the sequence of events that led to the Comet accidents (1954). Where did the fatigue crack initiate? What was the role of the window geometry?
- (b) Compute the stress concentration factor at the corner of a square window with a small fillet radius. Why did the original design use near-square windows?
- (c) What design changes were implemented after the investigation? How do these relate to the fatigue concepts studied in this course (stress concentration, SIF, residual strength)?

Problem 4.**Aloha Airlines Flight 243 case study**

- (a) Describe the concept of *multi-site damage* (MSD) and explain how it contributed to the Aloha Airlines fuselage failure (1988).
- (b) How does MSD differ from a single dominant crack in terms of residual strength and detectability?
- (c) What are the implications of MSD for damage-tolerance-based inspection programmes?

Problem 5.**Soderberg with K_t : safe design stress**

A cylindrical component with a circumferential notch ($K_t = 2.5$) is made of a steel with $\sigma_e = 600$ MPa, $\sigma_u = 900$ MPa and $\sigma_{f0} = 400$ MPa (for $R = -1$, unnotched). The notch sensitivity factor is $q = 0.8$.

- (a) Compute the fatigue strength reduction factor K_f .
- (b) Construct the Soderberg diagram for the notched component (replace σ_{f0} with σ_{f0}/K_f).
- (c) Determine the maximum allowable stress amplitude for a loading with $R = 0$ (pulsating).

Answer:  [Open in Colab](#)

11 Problems 11

Problem 1.

Fracture toughness and fracture assessment

An aircraft component is fabricated from an aluminium alloy with $K_{IC} = 30.2 \text{ MPa}\sqrt{\text{m}}$. The design specifies that, in the presence of an internal crack of total length $2a = 3 \text{ mm}$, fracture shall only occur for a tensile stress of 220 MPa.

- (a) Determine the geometry correction factor Y for this design condition (round to one decimal place).
- (b) Evaluate whether fracture will occur if a stress of 300 MPa is applied in the presence of an internal crack of total length $2a = 1 \text{ mm}$.

Problem 2.

Paris law integration

A CCT panel of width $W = 100 \text{ mm}$ is made from a carbon steel with $K_{IC} = 50 \text{ MPa}\sqrt{\text{m}}$. The panel is subjected to CA loading with $\Delta\sigma = 200 \text{ MPa}$ and $R = 0.25$. The initial crack has $2a_0 = 1 \text{ mm}$. The Paris law is:

$$\frac{da}{dN} = 1 \times 10^{-15} (\Delta K)^3 \quad (\text{mm/cycle, N} \cdot \text{mm}^{-3/2})$$

- (a) Compute $\sigma_{\max}$.
- (b) Describe the numerical procedure for computing N_f . What are the key steps?
- (c) Verify the order of magnitude of N_f by computing it for an infinite plate ($Y = 1$) using the closed-form Paris integral. Compare with the finite-width result $N_f \approx 5.26 \times 10^7$ cycles.

Problem 3.

Miner's rule with three blocks

An AISI 1045 steel component ($\sigma'_f = 1198.6 \text{ MPa}$, $b = -0.09$) is subjected to:

Block	σ_a (MPa)	n (cycles)
1	300 ($\sigma_m = 150$)	300 000
2	300 ($\sigma_m = 0$)	150 000
3	350 ($\sigma_m = 150$)	30 000

Using the modified Basquin relation $\sigma_a = (\sigma'_f - \sigma_m)(2N_f)^b$, compute:

- (a) $N_{f,i}$ for each block;
- (b) total Miner damage D ;
- (c) does the component survive?

Problem 4.**Soderberg diagram for $R = 0$**

A forged steel has $\sigma_e = 386$ MPa and $\sigma_{f0} = 296$ MPa (for $R = -1$).

- (a) Using the Soderberg relation, determine the fatigue limit (stress amplitude) for a pulsating loading ($R = 0$).
- (b) Compute $\sigma_{\max}$ for this condition.

Problem 5.**ASTM E399 validity**

A CT specimen ($B = 50$ mm, $W = 100$ mm, $a = 52$ mm) of a forged steel ($\sigma_e = 1050$ MPa) yields $K_Q = 156.8$ MPa $\sqrt{\text{m}}$.

- (a) whether the ASTM E399 validity conditions are satisfied for both the thickness B and the remaining ligament ($W - a$), where each must satisfy

$$B, W - a \geq 2.5 \left(\frac{K_Q}{\sigma_e} \right)^2$$

- (b) Is K_Q a valid K_{IC} ? If not, what is the maximum K_Q that would be valid with this specimen?

Problem 6.**Residual strength diagram**

For a CCT panel with $W = 200$ mm, made of an aluminium alloy with $K_{IC} = 37.2$ MPa $\sqrt{\text{m}}$ and $\sigma_e = 450$ MPa, consider a crack of total length $2a = 20$ mm. Determine:

- (a) the LEFM fracture stress using the Feddersen correction;
- (b) the net-section plastic collapse stress;
- (c) which failure mode governs for this crack size.

Problem 7.**S–N estimation and Soderberg with K_t**

Three aluminium alloys are candidates for a shaft of diameter d subjected to rotating bending with a superimposed static axial load. The shaft has a shoulder with $K_t = 2.5$ and $q = 0.85$. The loading gives $\sigma_a = 120$ MPa and $\sigma_m = 80$ MPa.

- (a) For each alloy, estimate σ_{f0} from σ_u (using $\sigma_{f0} \approx 0.35 \sigma_u$ for aluminium).
- (b) Compute $K_f = 1 + q(K_t - 1)$.
- (c) Apply the Soderberg criterion (with σ_{f0}/K_f and σ_e) to determine which alloys provide a safe design.

Alloy	σ_e (MPa)	σ_u (MPa)
Al 2024-T3	345	485
Al 6061-T6	275	310
Al 7075-T6	505	572

Answer:  [Open in Colab](#)

12 Problems 12

Formulary

Variable-amplitude loading and cycle counting.

For an irregular stress history, rainflow counting decomposes the sequence of reversals into fatigue cycles characterised by the stress range, stress amplitude and mean stress:

$$\Delta\sigma = \sigma_{\max} - \sigma_{\min}, \quad \sigma_a = \frac{\Delta\sigma}{2}, \quad \sigma_m = \frac{\sigma_{\max} + \sigma_{\min}}{2}.$$

Rainflow counting is commonly used before applying cumulative damage concepts to variable-amplitude loading spectra.

Basquin relation.

$$\sigma_a = \sigma'_f (2N_f)^b$$

where σ'_f is the fatigue strength coefficient and b is the fatigue strength exponent.

Palmgren–Miner linear damage rule.

$$D = \sum_{i=1}^k \frac{n_i}{N_{f,i}} \leq 1$$

where n_i is the number of cycles applied at block i and $N_{f,i}$ is the fatigue life under that constant-amplitude loading alone.

Modified Morrow strain–life relation.

$$\varepsilon_a = \frac{\sigma'_f - \sigma_m}{E} (2N_f)^b + \varepsilon'_f (2N_f)^c$$

where ε_a is the total strain amplitude, σ_m is the mean stress, E is Young's modulus, ε'_f is the fatigue ductility coefficient and c is the fatigue ductility exponent.

Smith–Watson–Topper (SWT) parameter.

$$\sigma_{\max} \varepsilon_a = \frac{(\sigma'_f)^2}{E} (2N_f)^{2b} + \sigma'_f \varepsilon'_f (2N_f)^{b+c}$$

where $\sigma_{\max}$ is the maximum stress in the cycle.

Fatemi–Socie critical-plane parameter.

$$FS = \frac{\Delta\gamma_{\max}}{2} \left(1 + k \frac{\sigma_{n,\max}}{\sigma_y} \right)$$

where $\Delta\gamma_{\max}/2$ is the maximum shear strain amplitude on the critical plane, $\sigma_{n,\max}$ is the maximum normal stress acting on that plane, σ_y is the yield stress and k is a material-dependent parameter.

Problem 1.**Rainflow counting for an irregular stress history**

A strain gauge mounted on an aircraft fuselage panel records the following stress-time history, expressed as a sequence of stress reversals in MPa:

$$0 \rightarrow 120 \rightarrow -40 \rightarrow 80 \rightarrow -100 \rightarrow 140 \rightarrow 20 \rightarrow -60 \rightarrow 0.$$

- (a) Apply the rainflow counting method to the sequence of reversals.
- (b) For each counted cycle or half-cycle, determine the stress range $\Delta\sigma$, stress amplitude σ_a , mean stress σ_m and cycle count.
- (c) Construct the corresponding rainflow matrix using stress range and mean stress bins.
- (d) Explain why rainflow counting is preferable to simple peak counting for fatigue damage assessment under variable-amplitude loading.
- (e) Implement the counting procedure in Python and present the results in tabular form.

Problem 2.**Miner damage under block loading**

An aluminium alloy component is subjected to the following repeated loading spectrum:

Block	$\Delta\sigma$ (MPa)	R	n (cycles)
1	420	0.1	2×10^5
2	360	0.1	8×10^5
3	480	0.1	5×10^4

The material follows the Basquin relation

$$\sigma_a = \sigma'_f (2N_f)^b$$

with

$$\sigma'_f = 950 \text{ MPa}, \quad b = -0.095.$$

- (a) Compute the stress amplitude σ_a for each block.
- (b) Determine the fatigue life $N_{f,i}$ associated with each block.
- (c) Compute the cumulative Miner damage $D = \sum n_i/N_{f,i}$.
- (d) Estimate the number of complete repeated spectra to failure.
- (e) Discuss the main limitations of Palmgren–Miner linear damage accumulation for aerospace structures subjected to overloads and underloads.

Problem 3.

Modified Morrow strain–life equation

A steel specimen is subjected to strain-controlled fatigue loading with total strain amplitude

$$\varepsilon_a = 0.006$$

and tensile mean stress

$$\sigma_m = 150 \text{ MPa.}$$

The material properties are:

$$\sigma'_f = 1050 \text{ MPa}, \quad \varepsilon'_f = 0.22, \quad b = -0.09, \quad c = -0.58, \quad E = 210 \text{ GPa.}$$

The modified Morrow relation is

$$\varepsilon_a = \frac{\sigma'_f - \sigma_m}{E} (2N_f)^b + \varepsilon'_f (2N_f)^c.$$

- Determine the fatigue life N_f numerically.
- Repeat the calculation for $\sigma_m = 0$.
- Quantify the influence of tensile mean stress on fatigue life.
- Plot the elastic and plastic strain-life contributions as a function of N_f .
- Explain why strain-life methods are particularly relevant for low-cycle fatigue.

Problem 4.

Smith–Watson–Topper mean-stress correction

A fatigue-critical location in an aircraft landing-gear component is subjected to a local cycle with

$$\sigma_{\max} = 420 \text{ MPa}, \quad \varepsilon_a = 0.0045.$$

The SWT parameter is defined as

$$SWT = \sigma_{\max} \varepsilon_a.$$

The corresponding strain-life equation is

$$\sigma_{\max} \varepsilon_a = \frac{(\sigma'_f)^2}{E} (2N_f)^{2b} + \sigma'_f \varepsilon'_f (2N_f)^{b+c}.$$

Use the following material properties:

$$\sigma'_f = 1100 \text{ MPa}, \quad \varepsilon'_f = 0.20, \quad b = -0.085, \quad c = -0.55, \quad E = 210 \text{ GPa.}$$

- Compute the SWT parameter.
- Determine the fatigue life N_f numerically.
- Compare the physical interpretation of SWT with the modified Morrow mean-stress correction.

-
- (d) Discuss why $\sigma_{\max}$ is an important variable in fatigue problems involving tensile mean stress.
 - (e) Implement the solution in Python using a root-finding procedure.

Problem 5.

Fatemi–Socie critical-plane parameter

A titanium alloy component is subjected to proportional combined tension–torsion loading. At a critical material point, the following quantities are obtained on the critical plane:

$$\gamma_a = 0.008, \quad \sigma_{n,\max} = 320 \text{ MPa}.$$

The material has yield stress

$$\sigma_y = 920 \text{ MPa},$$

and the Fatemi–Socie material parameter is assumed to be

$$k = 0.5.$$

The Fatemi–Socie parameter is

$$FS = \frac{\Delta\gamma_{\max}}{2} \left(1 + k \frac{\sigma_{n,\max}}{\sigma_y} \right).$$

- (a) Compute $\Delta\gamma_{\max}$ from the shear strain amplitude.
- (b) Compute the Fatemi–Socie parameter.
- (c) Explain the physical meaning of the normal-stress correction term.
- (d) Discuss why critical-plane methods are required for multiaxial fatigue assessment.
- (e) Compare, conceptually, Fatemi–Socie with a uniaxial strain-life approach.

Problem 6.

Conceptual questions on variable-amplitude and multiaxial fatigue

- (a) What is the main purpose of cycle counting in fatigue analysis?
- (b) Why can the sequence of loading cycles affect fatigue life even when the same set of cycles is applied?
- (c) What is the physical difference between low-cycle fatigue and high-cycle fatigue from the viewpoint of strain-life methods?
- (d) Why are mean-stress corrections needed in fatigue-life prediction?
- (e) In multiaxial fatigue, why may the maximum principal stress plane not coincide with the critical fatigue damage plane?
- (f) What additional difficulties arise under non-proportional multiaxial loading?

Answer:  [Open in Colab](#)

13 Assignments

Problem 1.

Consider a steel plate ($E = 210$ GPa, $\nu = 0.3$), as shown in Figure 13.1, with dimensions $2a \times 2b$ and thickness t ($t < a, b$). The plate has undergone a milling operation to create a circular hole with a nominal diameter of $2R$. A distributed tensile load along the x -axis of intensity σ_0 is applied on the faces of the plate parallel to the y -axis (Figure 13.1). Consider the following data: $2a = 500$ mm, $2b = 200$ mm, $t = 9$ mm, $2R = 20$ mm, $\sigma_0 = 300$ kN/m², and reply to the following tasks:

- Determine the strain and stress distributions in the region of interest and plot the strain and stress concentrations along the y -axis at $x = 0$. Solve the problem analytically, numerically, and experimentally.
- Empirically and numerically evaluate the effect of plate width and hole dimensions on the stress concentration factor K_t . Discuss the results.

Where applicable, report your developments and calculations in a structured code, implemented as a computational solution integrated into a Jupyter Notebook. For the numerical work, create the FE code to run the solution in batch mode. This data can be integrated with digital image correlation by creating a set of synthetic images of the test by deforming a reference speckle pattern using numerical displacement fields.

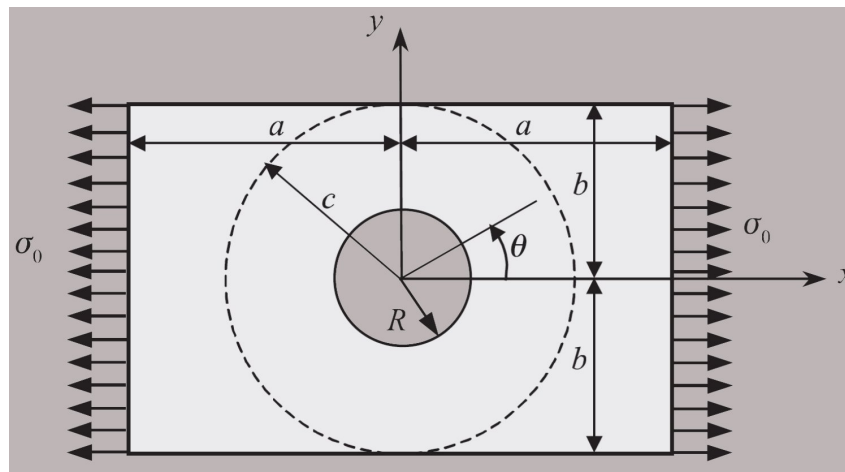


Figure 13.1: Problem 1.

Problem 2.

Consider the nominal geometry of the specimen presented in Figure 13.2. Carry out the experimental work by coupling the tensile tests under rupture with digital image correlation. Analyse the strain and stress concentration and evaluate the plastic zone development at the region of interest. Compare the experimental data with the numerical results obtained using the finite element method. Discuss the results.

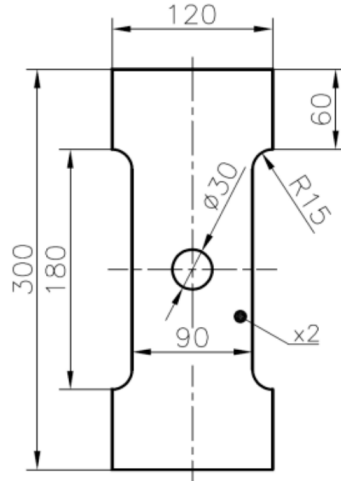


Figure 13.2: Problem 2.

Problem 3.

Consider a flat 2024-T3 aluminium alloy plate with a central circular hole of diameter $2R$. The plate is subjected to constant-amplitude fatigue (cyclic tensile) loading, perpendicular to the direction of two symmetric initial cracks emanating from the edge of the hole.

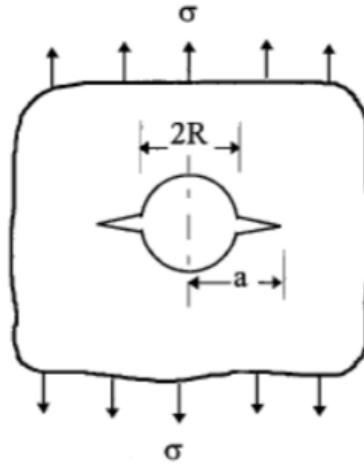


Figure 13.3: Problem 3.

The mode-I stress intensity factor for this geometry is

$$K_I = f\left(\frac{a}{R}\right) \cdot \sigma \sqrt{\pi a},$$

where a denotes the distance from the hole centre to the crack tip ($a \geq R$, with $a/R = 1$ corresponding to no crack), R is the hole radius, and $f(a/R)$ is the geometric correction factor tabulated in Table 13.1.

Table 13.1: Geometric correction factor $f(a/R)$ for a crack emanating from a hole in an infinite plate.

$\frac{a}{R}$	$f\left(\frac{a}{R}\right)$
1.01	0.3256
1.02	0.4514
1.04	0.6082
1.06	0.7104
1.08	0.7843
1.10	0.8400
1.20	0.9851
1.25	1.0168
1.30	1.0358
1.40	1.0495
1.80	1.0536

The 2024-T3 aluminium alloy has the following fatigue crack growth parameters:

- Paris law:
$$\frac{da}{dN} = C \cdot (\Delta K)^m$$

with $C = 1.0 \times 10^{-15}$ m/cycle and $m = 3.0$.

- Fracture toughness: $K_c = 30 \text{ MPa}\sqrt{\text{m}}$.
- Initial crack length: $a_0 = 1.01 R$.
- Young's modulus: $E = 73 \text{ GPa}$, Poisson ratio: $\nu = 0.33$.
- Applied cyclic stress: $\sigma_{\min}, \sigma_{\max}$ as per Table 13.2.

The work is differentiated across three groups (A, B, C). Each group adopts the geometry and loading combination specified in Table 13.2. The methodology and the deliverables are identical across groups; only the input parameters differ.

Table 13.2: Group-specific parameters for Problem 3.

Group	$2R$ (mm)	$\sigma_{\max}$ (MPa)	$\sigma_{\min}$ (MPa)	$\Delta\sigma$ (MPa)	R_σ
A	11.0	80	10	70	0.125
B	14.0	70	7	63	0.100
C	8.0	90	9	81	0.100

Tasks:

- Compute the evolution of the crack length $a(N)$ using the Paris law.
- Determine the critical crack length a_c such that $K_{\max} = K_c$.
- Estimate the total number of cycles N_f required for the crack to propagate from a_0 to a_c .
- Plot the curves $a(N)$ and $K_{\max}(a)$, and discuss the influence of $f(a/R)$ on the crack propagation, in particular in the early-crack regime.
- Discuss the implications for inspection intervals under a damage-tolerance philosophy: assuming a smallest detectable crack size $a_{\text{NDI}}/R = 1.10$ and a safety factor of 2 on the residual life, estimate the inspection interval N_{insp} .

Report your work in a structured Jupyter Notebook (template provided in Moodle), combining analytical and numerical solutions, with a final discussion that compares results across the three groups in the discussion class.

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